

INHA UNIVERSITY TASHKENT

FALL SEMESTER 2024

SOC 3010

OPERATING SYSTEM

CREDITS/HOURS PER WEEK : 3/3

COURSE TYPE : TECHNICAL CORE SEQUENCE

INSTRUCTOR

DR. A. R. NASEER

HEAD & PROFESSOR OF COMPUTER SCIENCE & ENGG

INHA UNIVERSITY TASHKENT

FALL SEMESTER 2024

SOC 3010

OPERATING SYSTEMS

CLASS SCHEDULE

GROUP1 ICE-1&2	TUE 11.00 – 12.30(A202)	THU 11.00 – 12.30(A202)
GROUP2 CSE-1	MON 09.30 – 11.00(A202)	THU 09.30 – 11.00(A202)
GROUP3 CSE-2	MON 11.00 – 12.30(A202)	WED 11.00 – 12.30(A202)
GROUP4 CSE-3	TUE 09.30 – 11.00(A202)	WED 09.30 – 11.00(A202)

OFFLINE CLASS RULES

TO BE FOLLOWED VERY STRICTLY

- 1. BE PRESENT IN THE CLASS ATLEAST 5 MINUTES BEFORE THE START OF THE CLASSES**
- 2. STUDENTS COMING LATE TO THE CLASSES AFTER 9.30AM & 11AM WILL NOT BE ALLOWED TO ENTER THE CLASS**
- 3. USE OF CELLPHONES & LAPTOPS STRICTLY PROHIBITED INSIDE THE CLASS**
- 4. CELL PHONES & LAPTOPS SHOULD NOT BE KEPT ON THE DESK**
- 5. CELLPHONES SHOULD BE KEPT EITHER IN YOUR POCKET OR BACKPACK/BAG YOU CARRY**
- 6. SLEEPING IN CLASS IS TOTALLY PROHIBITED**
- 7. BRING A NOTE BOOK FOR THIS COURSE AND MAKE NOTES IN THIS BOOK DURING THE CLASS**
- 8. NO DRINKING & NO EATING IS ALLOWED IN CLASS (PLEASE NOTE THAT CLASSROOM IS NOT CANTEEN)**

SOC 3010 – OPERATING SYSTEM

TEAM FORMATION INSTRUCTIONS FOR LAB ASSIGNMENTS

- **ALL LAB ASSIGNMENTS ARE TO BE COMPLETED IN GROUPS OF 4 TO 6 STUDENTS.**
- **YOU ARE REQUIRED TO FORM TEAMS COMPRISING A TEAM LEADER AND MEMBERS FOR CARRYING OUT LAB ASSIGNMENT1.**
- **A TEAM CAN HAVE A MAXIMUM OF 6 MEMBERS.**
- **A TEAM CAN HAVE MEMBERS DRAWN FROM OTHER SECTIONS ALSO.**
- **TEAM LEADER HAS TO PROVIDE THE TEAM DETAILS USING THE TEMPLATE PROVIDED.**
- **DULY-FILLED IN TEAM DETAILS MUST BE SUBMITTED BY EACH TEAM LEADER BY EMAIL TO MY OFFICIAL ACCOUNT n.abdulrahim@inha.uz BY FRIDAY, 13TH SEPTEMBER 2024.**
- **LATE SUBMISSIONS ARE NOT ENTERTAINED**
- **ADHERE TO THE DEADLINE STRICTLY**

INHA UNIVERSITY TASHKENT
FALL SEMESTER 2024

SOC 3010
OPERATING SYSTEMS

COURSE
SYLLABUS

SOC 3010

OPERATING SYSTEMS

COURSE DESCRIPTION

SOC 3010 OPERATING SYSTEMS

COURSE DESCRIPTION

This course is intended to provide the basic knowledge required to understand the inner-working of Computer Operating Systems.

This course will cover FIVE Basic Concepts of Operating Systems – Interrupt, Process, File, Memory, and I/O.

The students will read Linux kernel source code to see how these concepts are implemented. Based on this understanding, students will design a simple operating system.

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FALL SEMESTER 2024

SOC 3010
OPERATING SYSTEMS

COURSE LEARNING
OBJECTIVES & OUTCOMES



Course Learning Objectives

1. Understand basic concepts of operating system (Relation with Program Outcomes PO 2)
2. Implementing a simple operating system (Relation with Program Outcomes PO 3)

Program Outcomes (POs)

PO2 - Perform various experiments on Linux Kernel

(ABEEK PO2 : Analysis of experimental data- An ability to understand and analyze data, and plan and execute experiments)

PO3 - Design a Simple Operating System

(ABEEK PO3 : Problem formulation & solution - An ability to define, formulate and solve engineering problems)

SOC 3010

OPERATING SYSTEMS

PREREQUISITES :

[MANDATORY] DATA STRUCTURES

RECOMMENDED COURSES AFTER THIS COURSE:

COMPUTER SECURITY

COURSE SOFTWARE OR TOOL

LINUX (UBUNTU LINUX DISTRIBUTION)



Text Book

➤ **Understanding the Linux Kernel**

- **Daniel P/ Bovet and Marco Cesati**
Edition, O' Reilly Media 2005, ISBN 978-0596005658

Reference Books

➤ **Linux Kernel Development**

- **Robert Love**
Edition, Addison-Wesley 2010, ISBN 978-0672329463

➤ **Linux System Programming**

- **Robert Love**
Edition, O' Reilly Media 2013, ISBN 978-1449339531



Grading Policy

Methods of Evaluation :

Assessment will be made on the basis of one-
midterm Exam, One Final Examination, Weekly
Quizzes, Home Assignments and Term Project

Evaluation Criteria :

▪ Attendance	: 10%
▪ Weekly Quizzes	: 10%
▪ Home Assignment	: 10%
▪ Term Project Work	: 20%
▪ Mid-Term Exam	: 25%
▪ Final Exam	: 25%

Total : 100%

▪ **GRADES AWARDED : A+, A0, B+,B0, C+,C0, D+,D0, F (FAIL)**

SOC 3010 OPERATING SYSTEMS

Course Topics & Weekly Lecture Breakdown

I WEEK

Lecture Topic : Overview (Ch 1)

Basic Operating System Concepts, Linux Versus Other Unix-Like Kernels, Hardware Dependency, Linux Versions, An Overview of the Unix Filesystem, An Overview of Unix Kernels

Homework Assignment

Understand the usage of linux commands such as ls, cp, mv, etc., and system calls such as read(), write(), etc.

Practise a given list of basic linux commands and system calls

SOC3010 OPERATING SYSTEMS

Course Topics & Weekly Lecture Breakdown

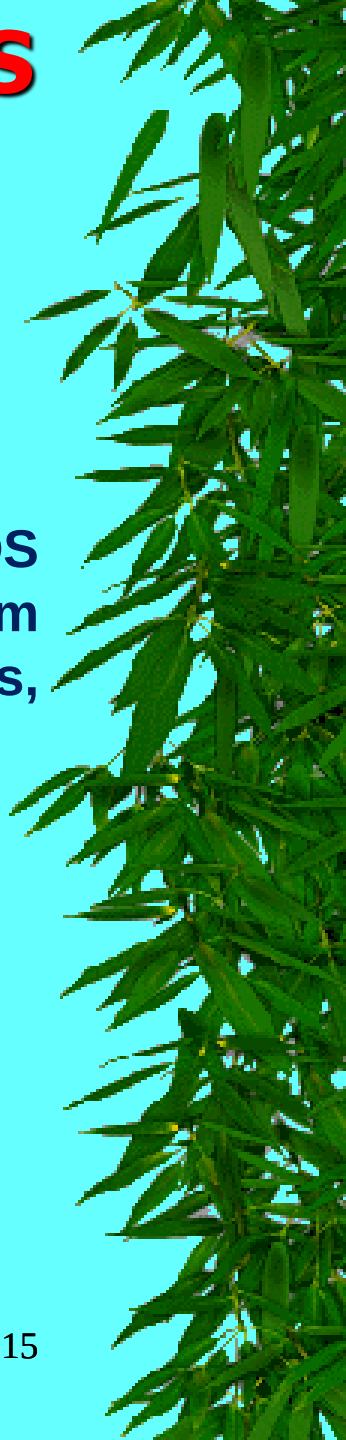
II WEEK

Lecture Topic : Linux kernel (Ch 1, Appendix A)

An Overview of Unix Kernels, System Startup – BIOS Bootstrapping procedure, Boot Loader, Booting Linux from Hard disk (LILO – Linux LOader), setup and startup functions, Shell Programming

Homework Assignment

- Installation of linux with kernel source.**
- Explain the kernel code structure.**
- Show the basic techniques to read through the linux kernel**
- Writing Shell Scripts**



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Course Topics & Weekly Lecture Breakdown

III WEEK

Lecture Topic : Memory Addressing (Ch 2)

Memory Addresses- logical, linear, physical addresses, Segmentation in Hardware - Segmentation Registers, Segmentation Descriptors, Global Descriptor table (GDT), Local Descriptor Table(LDT), Segment selector, Segmentation in Linux, Paging in Hardware, Paging in Linux

Home Assignment : Compiling linux (Linux kernel)

- How to compile the Linux kernel source code.
- How to make some modification and recompile the kernel.

TERM PROJECT TITLE SELECTION & SUBMISSION FOR APPROVAL

SOC 3010 OPERATING SYSTEMS

Course Topics & Weekly Lecture Breakdown

IV WEEK

Lecture Topic : Interrupt concept (Ch 4)

Explain basic interrupt concept, Role of Interrupt Signals, Type of Interrupts, Types of Exceptions, Interrupt and Exception vectors, IRQs, Advanced Programmable interrupt controller(APIC), Multi-APIC System, Interrupt Descriptor Table(IDT), Hardware Handling of Interrupts and Exceptions

Home Assignment

Show a detailed example with keyboard interrupt
Explain kernel interrupt code for keyboard interrupt.

V WEEK

Lecture Topic : Writing a sample Linux Kernel Module

Utilities to Manipulate Kernel Module, Creating Makefile to compile kernel module, Inserting the sample kernel module, remove the sample kernel module

Interrupt example: Keyboard interrupt (Linux kernel)

Home Assignment

Explain how to change kernel keyboard interrupt handling code to modify keyboard interrupt behavior in Linux.

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Course Topics & Weekly Lecture Breakdown

VI WEEK

Lecture Topic : Process: basic data structure (Ch 3)

Explain basic concept of process and basic data structure to handle process.

Explain what information the OS should keep in the process control block to manage processes.

Home Assignment

- Find process related data structure in the kernel code.

TERM PROJECT PROPOSAL SUBMISSION

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Course Topics & Weekly Lecture Breakdown

VII WEEK

MID-TERM EXAM

(21-10-2024 to 25-10-2024)



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Course Topics & Weekly Lecture Breakdown

VIII WEEK

Lecture Topic : Process: system calls (Ch 3) & Socket Programming

Explain the inner working of process related system calls: fork, exec, exit, wait.

Inter-Process Communication (IPC) - Socket Programming

Home Assignment

-Write example code that uses the system calls fork, exec, exit, wait.

-Read kernel code related to process.

-Socket Programming demonstrating Inter Process Communication

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Course Topics & Weekly Lecture Breakdown

IX WEEK

Lecture Topic : Inter Process communication- IPC (Ch19)

**Basic Mechanisms to allow interprocess communication -
Pipes and FIFOs (named pipes), Semaphores, Messages,
Shared memory regions, Sockets**

**System V IPC Resources- IPC Semaphores, IPC Messages,
IPC Shared Memory, POSIX Message Queues**

**Term Project work – Using Inter-process
Communication**

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Course Topics & Weekly Lecture Breakdown

X WEEK

Lecture Topic : File: on-disk file system (Ch 18)

Linux Disk File system based on Inode Structure – Extended File System(Ext), General Characteristics of Extended File System Ex2, Ex3, Data Structures, Ext File System, File Types, directory.

Reading Assignment

- Read kernel code for disk file systems**

Term Project Work Continued

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Course Topics & Weekly Lecture Breakdown

XI WEEK

Lecture Topic : File: on-disk file system (Ch 18)

Ext Memory Data Structure, Creating Ext File System, Data Blocks Addressing, Journaling File Systems.

Reading Assignment

- Read kernel code for disk file systems

Term Project Work Continued



XII WEEK

Lecture Topic : File: on-memory file system & File system calls (Ch 12, Ch 16)

Virtual File System(VFS), Common File Model, System Calls handled by the VFS, VFS Data Structures, Superblock operations, Inode Objects, Inode operations, File Objects, File Operations, dentry Objects, Files associated with a process, Special File Systems, Mounting and Unmounting a file system, Implementation of VFS System Calls, File Locking- File handling system calls

Reading Assignment

- Read the kernel code and find the related code in the kernel that handles the above tasks including file handling system calls

Term Project Work Continued

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Course Topics & Weekly Lecture Breakdown

XIII WEEK

Lecture Topic : Memory: paging (Ch 8)

Memory Management – Page Frame Management, Page Descriptors, Non-Uniform Memory access, Memory zones, Zoned Page Frame Allocator, Buddy System Algorithm, Zone Allocator, Memory Area Management, Slab Allocator, Non-Contiguous Memory Area Management.

- Term Project Work Continued

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Course Topics & Weekly Lecture Breakdown

XIV WEEK

Lecture Topic : Memory: paging data structure, page fault (Ch 9) and System Calls & Signals (Ch 10 & 11)

Process address Space, Memory Descriptor, Memory regions, Memory Region Data Structures, Memory Region Handling page fault exception, and the related kernel code to handle paging and page fault.

POSIX API vs System Calls, System Call handler and Service Routines, Entering and Exiting a System Call, parameter Passing

Compare signal vs interrupt. Explain various kinds of signal. Actions Performed upon Delivering a Signal, Catching a Signal, System Calls Related to Signal Handling, Related data structure and kernel code to handle signal.

- Term Project Work Continued

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Course Topics & Weekly Lecture Breakdown

XV WEEK

FINAL EXAM



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Course Topics & Weekly Lecture Breakdown

XVI WEEK

23rd to 28th December 2024

EVERY DAY FROM MORNING 9AM TO EVENING 7PM

- Term Project Presentation & Demo**
- Term Project Report Submission**

SOC 3010 OPERATING SYSTEM

TERM PROJECT GUIDELINES

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Term Project Guidelines

- ★ **Term Project involves the following main tasks :**
 - **Selection of a particular IPC related problem**
 - **Project Design & Implementation/Simulation**
 - **Preparation of the Project Report**
 - **Presentation of the Project work**
- **Each Project Team can consist of 4 to 8 members (or more depending on the size of the section) with a maximum of 10 project teams in a section.**

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Term Project Guidelines

- Project Teams are given time till **3rd week of the semester** to select a IPC (Inter-Process Communication) related to real-time problem pertaining to society /social networking (sectors include Banking, Airline/Railways/Bus Transports, Education/University, Hotel/Restaurant, HealthCare, COVID-19 Contact tracing/Management, Online Education, Traffic Control/Management, Digital Money Transfer/ Digital Asset Management, Social media or any other domain of interest) and carry out the Project work in a team.
- Each project team is required to develop the software that model the business process according to the System Request. It must be implemented using client – server architecture, all inter process communications should be organized using the popular IPC mechanism -Sockets.

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Term Project Guidelines

- The title of the selected Term Project along with abstract, Brief project description, Requirements definition document in the form of a Project Preliminary Document also known as Project Proposal must be submitted by the **end of 6th week of the semester.**
- No two Teams can have the same Project. The problem selected must belong to different areas.
- Project Presentation will be held in the **16th week of the semester (23rd to 28th December 2024)**

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Term Project Guidelines



- **Maximum duration of the presentation will be 45 minutes followed by 15 minutes for question/answer session for each Project team. All members of the team must present a part of the Project carried-out in rotation. The time-division for presentation among the members must be done judiciously so that complete Project work carried out is presented in a Perfect way without missing any detail and after that a live Demo of the Project developed will be given – Demo duration : 30 minutes.**
- **The Final Project Report must include the following Sections : Abstract, Introduction, Project Overview, Requirements definition section including Use case diagram, Activity diagrams, Class diagrams, Behavioral diagram, Project Design & Implementation, Simulation Results & Discussions, Conclusion, Future Work, References.**
- **The Project Report must be submitted atleast one day before the Presentation or on the day of Presentation before the Presentation.**
- **Project Deliverables : Project Report(Soft Copy) with Simulation Results, Source Code developed, Presentation PPTs**

Contact Details :

E-mail : n.abdulrahim@inha.uz
dr_arnaseer@hotmail.com

Room No. : 402/4

Office Hours :

WEDNESDAY : 2PM TO 4PM

THURSDAY : 2PM TO 4 PM

Attendance Policy

➤ **All Regular students should attend all classes and laboratory sessions.**

➤ **Students who are absent without valid reason for more than 25% the assigned hours (less than 75% Attendance) for the course will given a Grade “F”**

Academic Dishonesty Policy

➤ In order for instructors to fairly assess the quality and quantity of a student's learning (through course grades) as determined by work that students represent as their own, a relationship of trust between instructor and student is essential. Because violations of academic integrity most often involve, but are not limited to, efforts to deceive instructors, they represent a breach of the trust relationship between instructor and student, and undermine the core values of the university.

➤ For these reasons, the University and its instructors treat issues of academic dishonesty as serious violations of academic trust, and conduct rigorous investigations of students suspected of committing such acts.

ACTS OF ACADEMIC DISHONESTY INCLUDE, BUT ARE NOT LIMITED TO THE FOLLOWING:

- the illegitimate use of materials in any form during a quiz or examination
- copying answers from the quiz or examination paper of another student
- plagiarizing (submitting as one's own ideas the work of another) or falsifying materials or information used in the completion of any assignment which is graded or evaluated as the student's individual effort
- submitting the same work for more than one course without the consent of the instructors of each course in which the work is submitted
- copying material from a web page and submitting it as one's own work
- quoting extensively from a document without making proper references to the source

☐ If a student is found committing such acts in a quiz or home assignment or exam or term paper or term project he/she will be given a grade 0 in that part of the course.

SOC 3010

OPERATING SYSTEM

END OF

WEEK 1

DISCUSSION ON

COURSE SYLLABUS

WISH YOU ALL THE BEST

INHA UNIVERSITY TASHKENT

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